

Modeling Cyberphysical Systems

Sayan Mitra

Verifying cyberphysical systems

mitras@illinois.edu

Outline

- Modeling cyber-physical systems
 - Hybrid automata
 - Executions
 - Urgency
- Zeno
- Hybrid stability

Why combine discrete and continuous time models?

Engineering systems can have dynamics with widely different time constants

- Software for AV updates at 1 GHz --- human scale motion 10 Hz
- Electrical circuit breakers 50 Hz --- power system dynamics 0.1 – 1 Hz

Widely different time constants can make computation numerically unstable

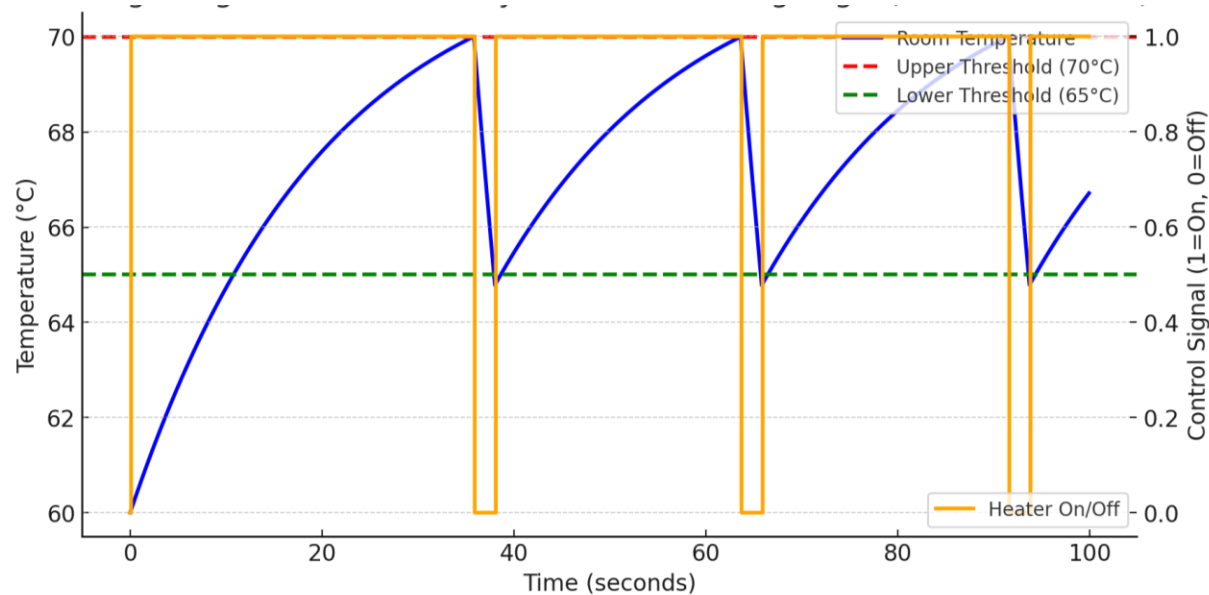
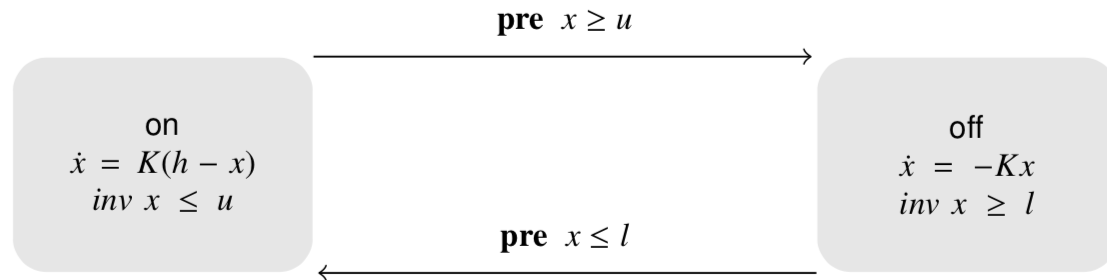
Some state changes are naturally described by automata:

- programs, automata, SMT, switching rules, etc.

while others are naturally described by ODEs:

- motion of planets and satellites, physical laws, weather

Example. Thermostat automaton



automaton Thermostat($u, l, K, h : \text{Real}$) where $u > l$

type Status enumeration [*on*, *off*]

actions

turnOn; turnOff;

variables

$x : \text{Real} := l$ loc: Status := on

transitions

turnOn

pre $x \leq l \wedge \text{loc} = \text{off}$

eff loc := on

trajectories

modeOn

evolve $d(x) = K(h - x)$

invariant loc = on $\wedge x \leq u$

turnOff

pre $x \geq u \wedge \text{loc} = \text{on}$

eff loc := off

modeOff

evolve $d(x) = -Kx$

invariant loc = off $\wedge x \geq l$

Determinism vs nondeterminism

mode invariants

Recall language for automata

An **automaton** tuple $\mathcal{A} = \langle X, \Theta, A, \mathcal{D} \rangle$

- X is a set of variables; each $x \in X$ is associated with a type, $type(x)$
 - A **valuation** for X maps each variable in X to its type
 - Set of all valuations: $val(X)$ is the **state space**
- $\Theta \subseteq val(X)$ set of **initial** or **start states**
- A is a set of names of **actions** or **labels**
- $\mathcal{D} \subseteq val(X) \times A \times val(X)$ is the set of **transitions**
 - a transition is a triple (u, a, u')
 - defined by **pre** and **eff**
 - We write it as $u \rightarrow_a u'$

```
automaton DijkstraTR(N:Nat, K:Nat), where K > N
type ID: enumeration [0,...,N-1]
type Val: enumeration [0,...,K]
actions
  update(i:ID)
variables
  x:[ID -> Val]
transitions
  update(i:ID)
    pre i = 0 /\ x[i] = x[N-1]
    eff x[i] := (x[i] + 1) % K

  update(i:ID)
    pre i > 0 /\ x[i] ~= x[i-1]
    eff x[i] := x[i-1]
```

Trajectories

For set of variables X and time interval J of the form $[0, T]$, $[0, T)$ or $[0, \infty)$, a **trajectory** for X is a function $\tau: J \rightarrow \text{val}(X)$

We will specify τ as *solutions of* differential equations

first state of τ . $fstate := \tau(0)$

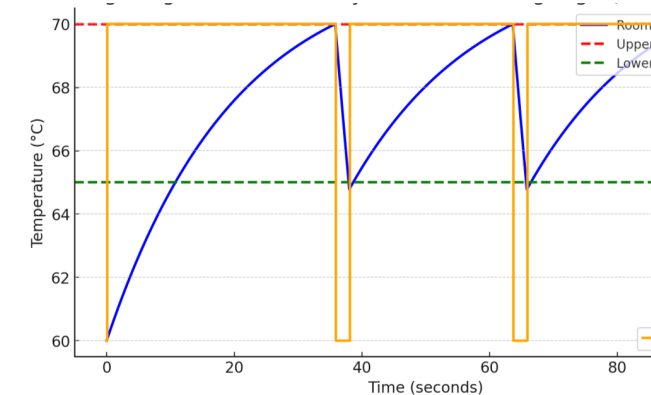
If τ is right closed then the **limit state** of τ . $lstate = \tau(T)$

If τ is finite then **duration** of τ is $\tau.dur = T$

Domain of τ . $dom = J$

A **point trajectory** is a trajectory with $\tau.dur = [0,0]$

A **prefix** τ' of $\tau: [0, T] \rightarrow \text{val}(X)$, is a function $\tau': [0, T'] \rightarrow \text{val}(X)$ such that $T' \leq T$ and $\tau'(t) = \tau(t)$ for all $t \in \tau'.dom$



Hybrid Automaton

$$\mathcal{A} = (X, \Theta, A, \mathcal{D}, \mathcal{T})$$

- X : set of **state variables**
 - $Q \subseteq \text{val}(X)$ set of **states**
- $\Theta \subseteq Q$ set of **start states**
- set of **actions**, $A = E \cup H$
- $\mathcal{D} \subseteq Q \times A \times Q$
- $\mathcal{T}$: set of **trajectories** for X which is closed under **prefix, suffix, and concatenation**

Semantics: Executions and Traces

An **execution** of $\mathcal{A}$ is an (possibly infinite) alternating sequence $\alpha = \tau_0 a_1 \tau_1 a_2 \tau_2 \dots$ where

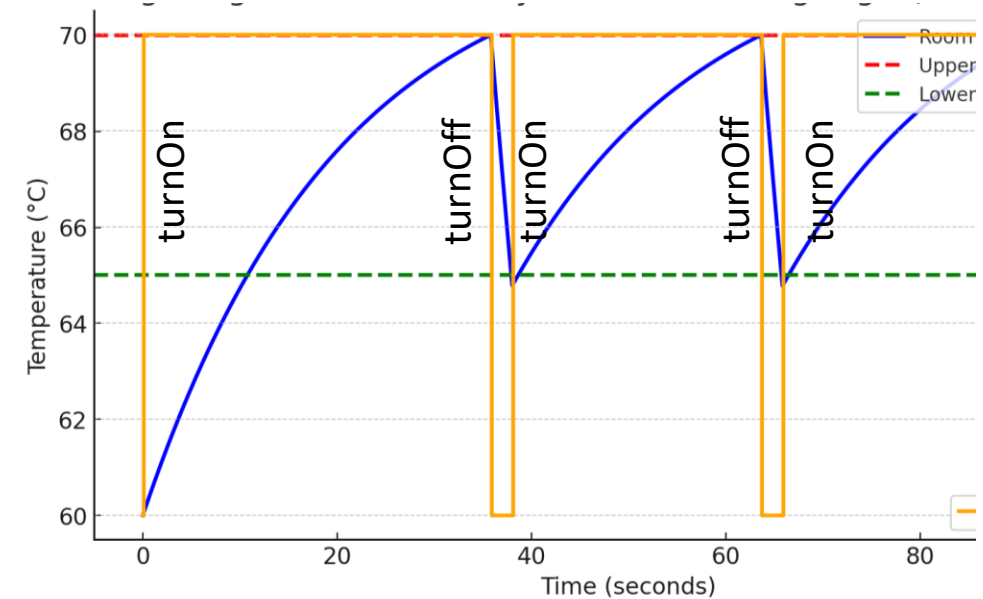
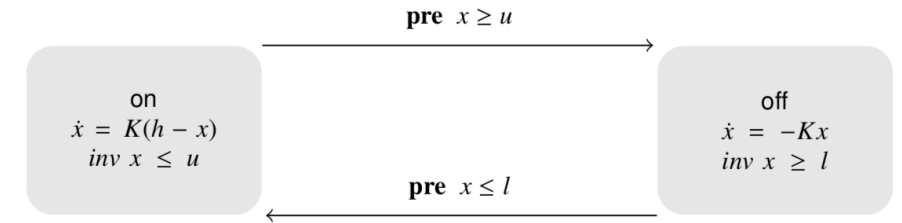
- $\forall i, \tau_i.lstate \xrightarrow{a_{i+1}} \tau_{i+1}.fstate$
- If $\tau_0.fstate \in \Theta$ then α is an **execution**

Execs $_{\mathcal{A}}$ set of all executions

First state of α is $\alpha.fstate = \tau_0.fstate$

If α is **finite and closed** $\tau_0 a_1 \tau_1 a_2 \tau_2 \dots \tau_k$ then $\alpha.lstate = \tau_k.lstate$

A state x is *reachable* if there exists an execution α with $\alpha.lstate = x$



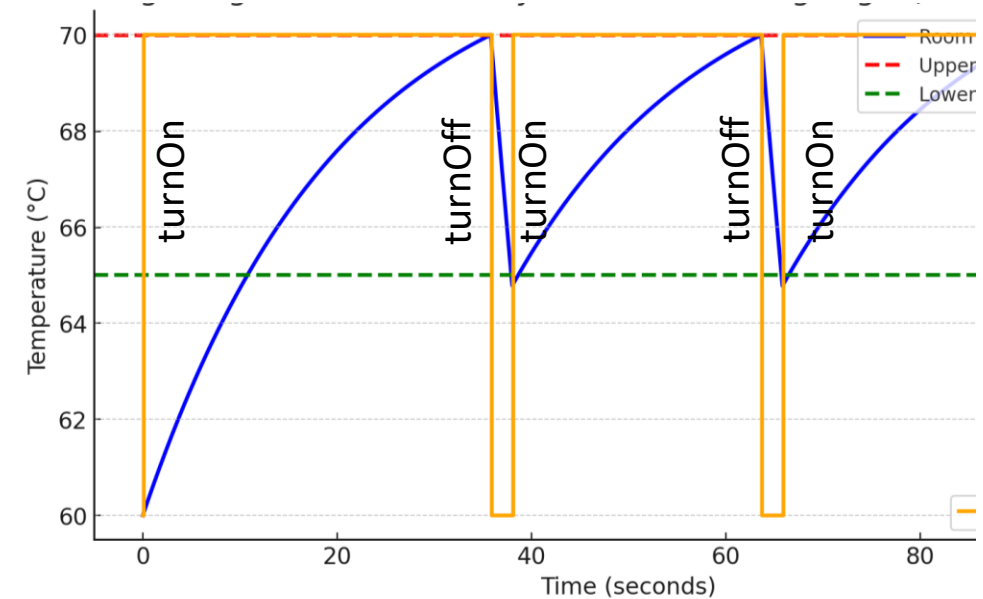
Mode invariants: Urgency in transitions

The trajectories in a hybrid automaton are often specified as solutions of ODEs

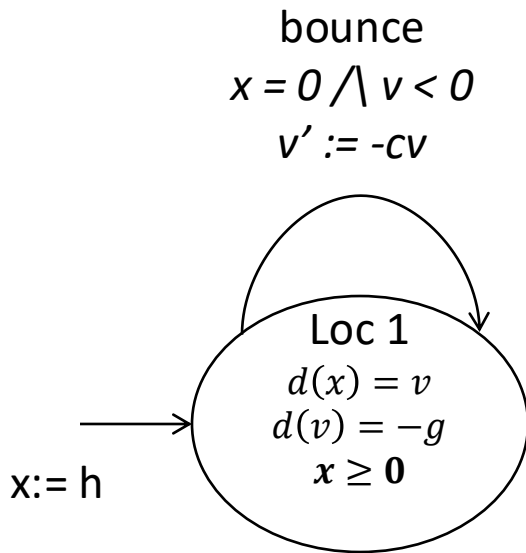
- $\dot{x} = f(x)$
- $\dot{x} = Ax$
- $\dot{x} = c$

Solutions of ODEs $\tau(x_0, t)$ define a family of trajectories which are closed under prefix, suffix, and concatenation

Nothing prevents a trajectory to continue forever unless an action is forced to occur by a **mode invariant**



Example. Bouncing Ball



Graphical Representation used in many articles

automaton Bouncingball(c, h, g)

variables: x : Reals := h , v : Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

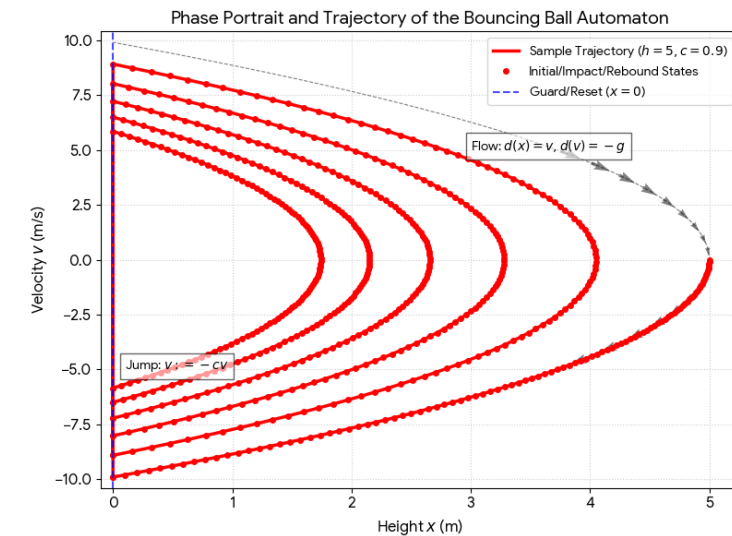
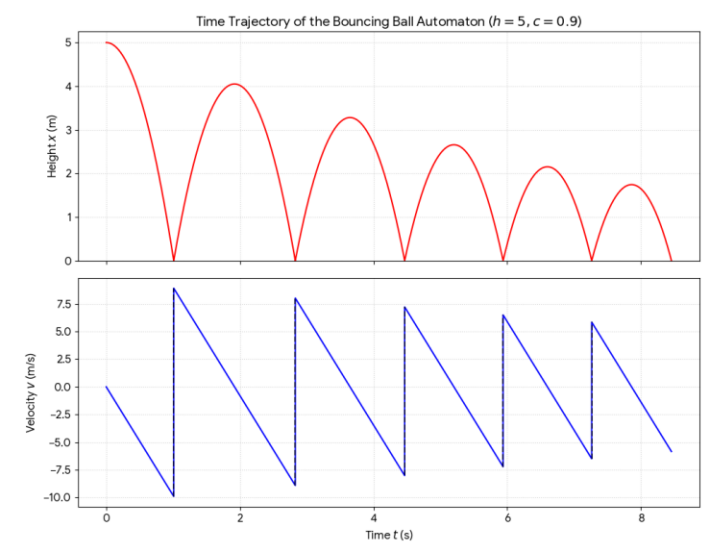
eff $v := -cv$

trajectories:

Loc1

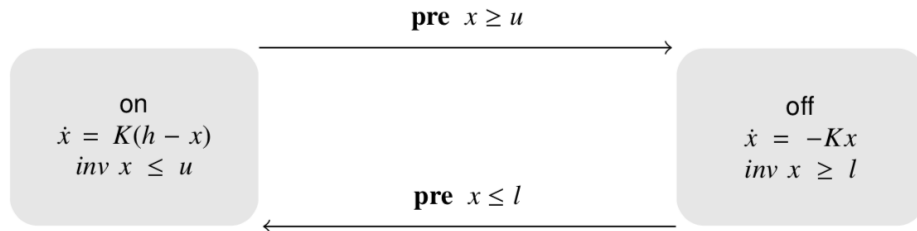
evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$



mode invariant, not to be confused with invariants of the automaton

Urgency: Must transitions



An **urgent** transition (or action) is an action that has to occur as soon as it is enabled

Introduce special syntax for creating urgent transitions, but we can use mode invariants to accomplish this

automaton Thermostat($u, l, K, h, d : \text{Real}$) where $u > l$

type *Status* enumeration [*on*, *off*]

actions

turnOn; turnOff;

variables

$x : \text{Real} := l$ $loc : \text{Status} := on$

transitions

turnOn

pre $x \leq l \wedge loc = off$

eff $loc := on$

turnOff

pre $x \geq u \wedge loc = on$

eff $loc := off$

trajectories

modeOn

evolve $d(x) = K(h - x)$

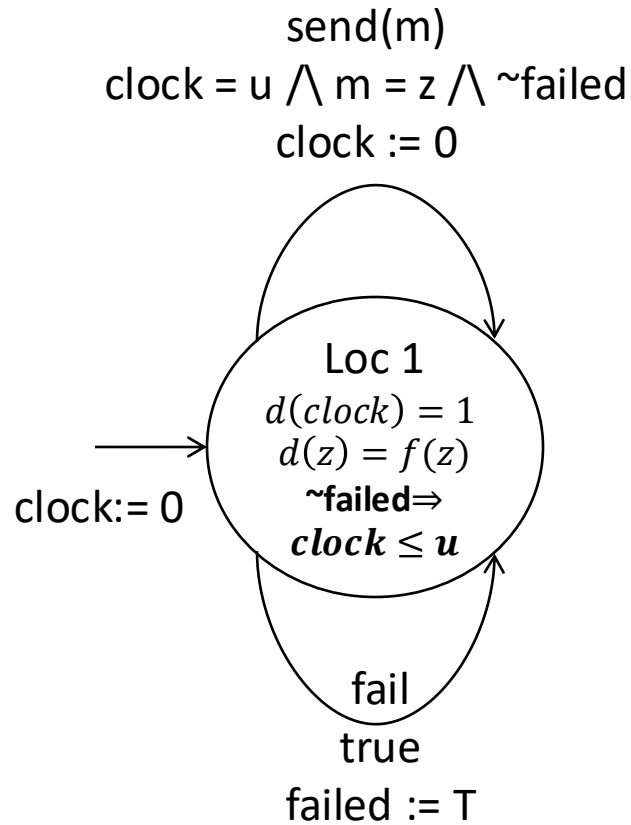
invariant $loc = on \wedge x \leq u + d$

modeOff

evolve $d(x) = -Kx$

invariant $loc = off \wedge x \geq l - d$

Another Example: Periodically Sending Process



Automaton PeriodicSend(u)

variables: analog

$\text{clock} : \text{Reals} := 0, z : \text{Reals}, \text{failed} : \text{Boolean} := F$

actions: $\text{send}(m : \text{Reals}), \text{fail}$

transitions:

$\text{send}(m)$

pre $\text{clock} = u \wedge m = z \wedge \sim \text{failed}$

eff $\text{clock} := 0$

fail

pre true

eff $\text{failed} := T$

trajectories:

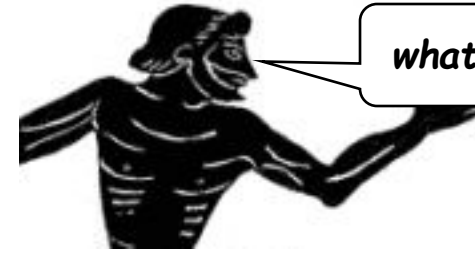
evolve $d(\text{clock}) = 1, d(z) = f(z)$

invariant $\text{failed} \vee \text{clock} \leq u$

Zeno's Paradox

Achilles, the fastest athlete, greatest warrior

Zeno, Greek philosopher

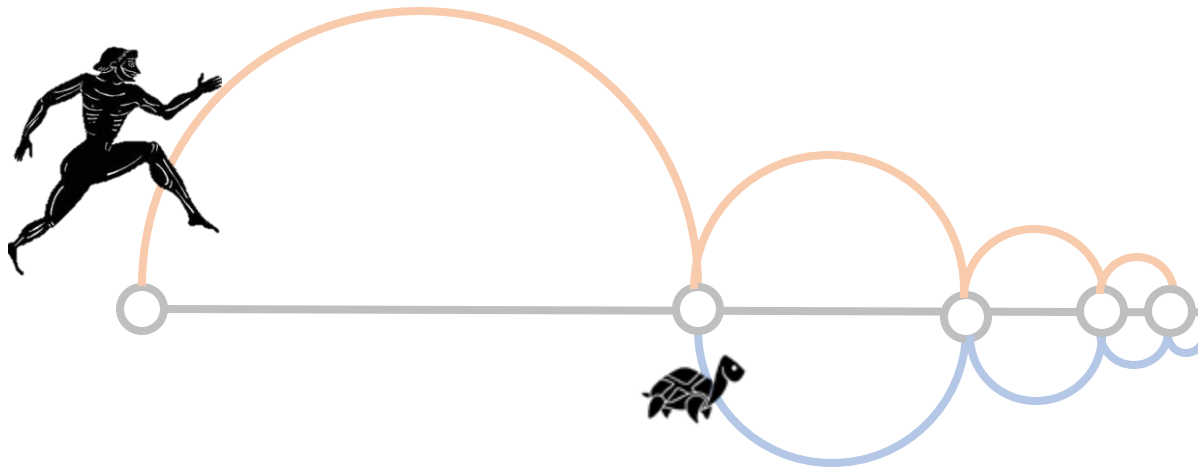


whatever!

You couldn't even beat a turtle



Achilles runs 10 times faster than the tortoise, but the tortoise gets to start 1 second earlier. Can Achilles ever catch Turtle?



After $1/10^{\text{th}}$ of a second, Achilles reaches where the Turtle (T) started, and T has a head start of $1/10^{\text{th}}$ second.

After another $1/100^{\text{th}}$ of a second, A catches up to where T was at $t=1/10$ sec, but T has a head start of $1/100^{\text{th}}$

...

T is always ahead ...

Lesson: Mixing discrete transitions with continuous motion can be tricky!

Defining stability of hybrid automata

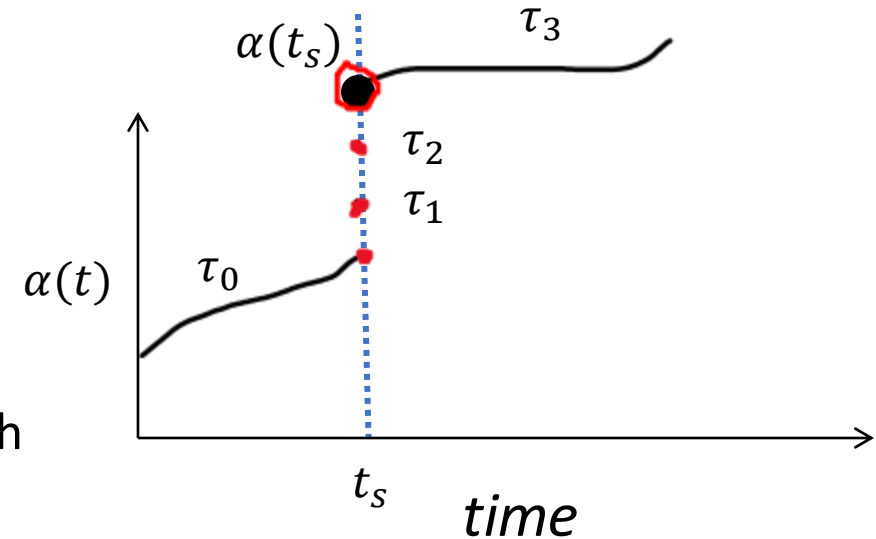
Given an admissible execution $\alpha = \tau_0 a_1 \tau_1 a_2 \tau_2 \dots$

Recall, stability is defined in terms of $|\alpha(t)|$; two issues

- We would like to view an execution as $\alpha: [0, \infty) \rightarrow \text{val}(X)$ But, how to define $\alpha(t)$?
- define $\alpha(t_s) = \alpha'.lstate$ where α' is the longest **prefix** of α with $\alpha'.ltime = t_s$
- Also, $\alpha(t)$ has valuations of discrete variables (loc) and continuous variable (x): ignore discrete variable when discussing stability $\alpha(t) \equiv \alpha(t)[x]$

Now, we can import all the definitions of Lyapunov stability and asymptotic stability from dynamical system models to hybrid automata

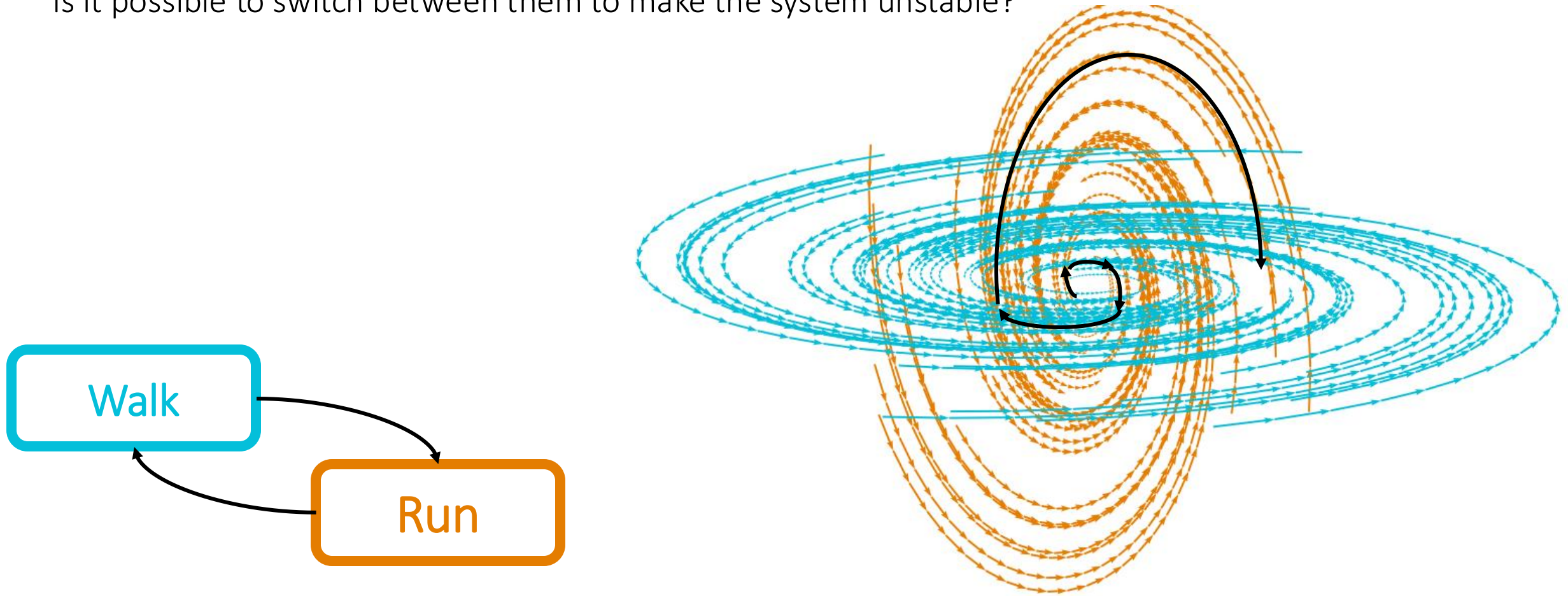
Question. How can we check stability of a hybrid automaton?



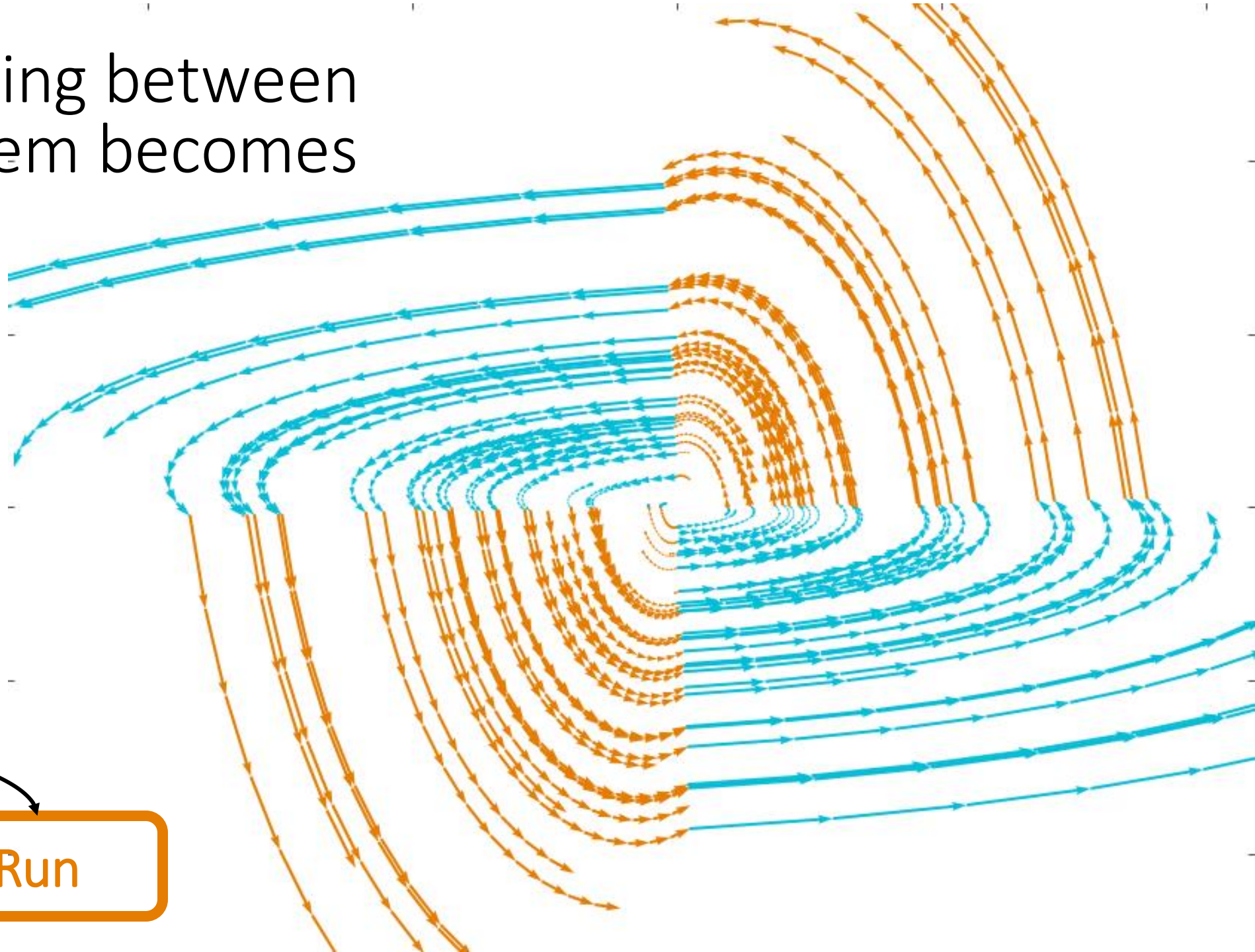
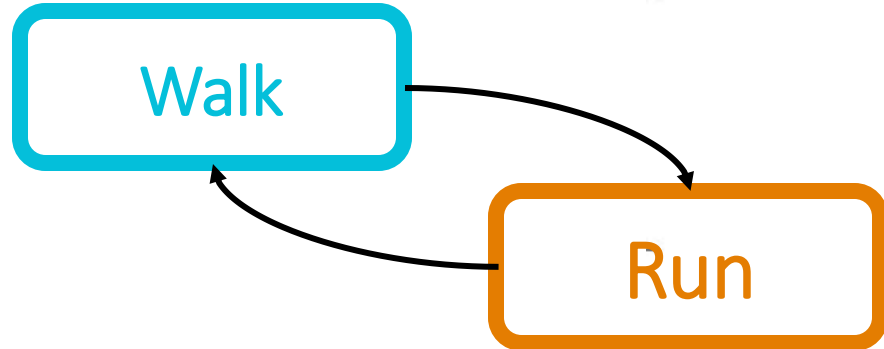
Stability of a hybrid automaton

Each of the modes of a walking robot are asymptotically stable

Is it possible to switch between them to make the system unstable?



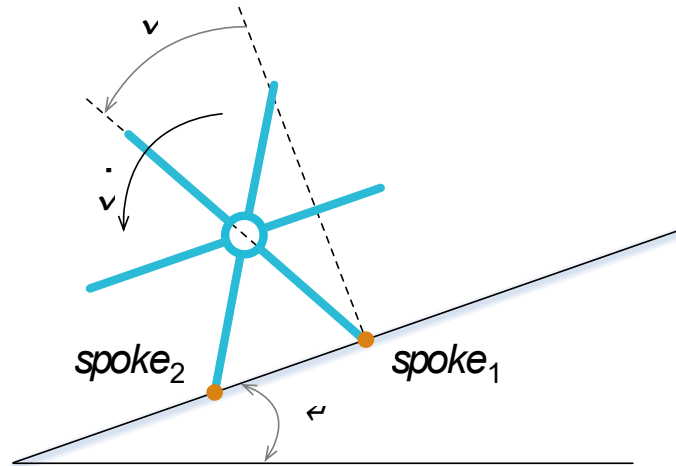
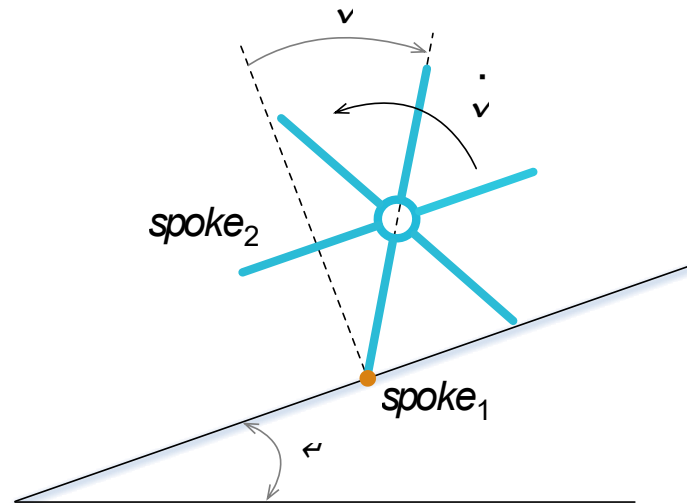
Yes! By switching between them the system becomes unstable



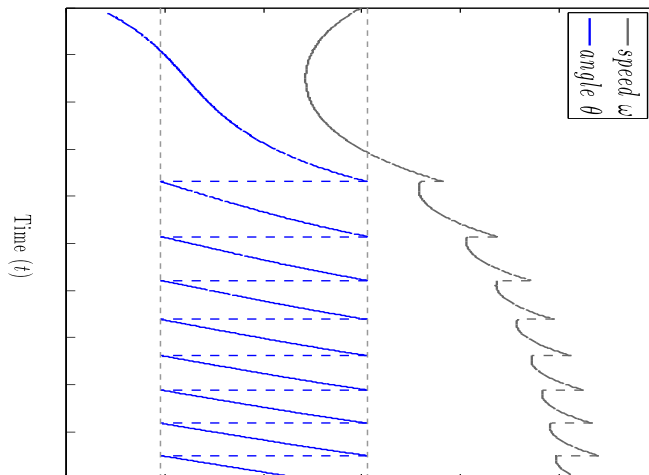
Special kinds of executions

- **Infinite:** Infinite sequence of transitions and trajectories
 $\tau_0 \ a_1 \ \tau_1 a_2 \tau_2 \ \dots$
- **Closed:** Finite with final trajectory with closed domain
 $\tau_0 \ a_1 \ \tau_1 a_2 \tau_2 \ \dots \tau_k$ and $\tau_k \cdot dom = [0, T]$
- **Admissible:** Infinite duration
 - May or may not be infinite
 - $\tau_0 \ a_1 \ \tau_1 a_2 \tau_2 \ \dots$
 - $\tau_0 \ a_1 \ \tau_1 a_2 \tau_2 \ \dots \tau_k$ with $\tau_k \cdot dom = [0, \infty)$
- **Zeno:** Infinite but not admissible
 - Infinite number of transitions in finite time

Rimless wheel



θ, ω



automaton RimlessWheel(α, μ : Real, n : Nat)

const β : Real := $2 \pi / n$

type Spokes: enumeration [1,...,n]

actions

impact

variables

pivot: Spokes := 1

θ : Real := 0

ω : Real := 0

transitions

impact

pre $\theta \geq \beta / 2$

eff pivot := pivot + 1 mod n

$\theta := \beta / 2$

$\omega := \mu \omega$

trajectories

swing

evolve

$d(\theta) = \omega$

$d(\omega) = \sin(\theta + \alpha)$

invariant $\theta \leq \frac{\beta}{2}$

Invariants and reachability

A state x of $\mathcal{A}$ is **reachable** if $\exists$ an execution α with $\alpha.lstate = x$

$Reach_{\mathcal{A}}(\Theta)$ all reachable state from Θ

$Reach_{\mathcal{A}}(\Theta, T)$ states reachable within time T

$Reach_{\mathcal{A}}(\Theta, k)$ states reachable within k transitions

$Reach_{\mathcal{A}}(\Theta, T, k)$ states reachable up to time k transitions and time T

An invariant $I \subseteq val(X)$ is a set of states that contains $Reach_{\mathcal{A}}(\Theta)$

How to prove invariants of hybrid automata

Recall the theorem we used for proving I is an invariant for automaton $\mathcal{A} = \langle X, \Theta, A, D \rangle$

Theorem 7.1. Given an automaton $\mathcal{A} = \langle X, \Theta, A, \mathbf{D} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta$, $x \in I$ and
- (Transition closure) For any action $a \in A$, if and $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and

Then I is an inductive invariant of $\mathcal{A}$.

How to prove invariants of hybrid automata

Theorem 7.1. Given an HIOA $\mathcal{A} = \langle X, \Theta, A, \mathbf{D}, \mathbf{T} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta$, $x \in I$ and
- (Transition closure) For any action $a \in A$, if $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and
- (Trajectory closure) For any trajectory $\tau \in \mathbf{T}$ if $\tau.fstate \in I$ then $\tau.lstate \in I$

Then I is an inductive invariant of $\mathcal{A}$.

How to prove invariants of hybrid automata

Theorem 7.1. Given an HIOA $\mathcal{A} = \langle X, \Theta, A, \mathbf{D}, \mathbf{T} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta, x \in I$ and
- (Transition closure) For any action $a \in A$, if $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and
- (Trajectory closure) For any trajectory $\tau \in \mathbf{T}$ if $\tau.fstate \in I$ then $\tau.lstate \in I$

Then I is an inductive invariant of $\mathcal{A}$.

Proof. Consider any reachable state $x \in \text{Reach}_{\mathcal{A}}$. By the definition of a reachable state, there exists an execution α of $\mathcal{A}$ with $\alpha.lstate = x$. We proceed by induction on the length of the execution α . For the base case, α consists of a single starting state $x \in \Theta$, and, by the *start condition*, $x \in I$. For the inductive step, we consider two subcases:

Case 1: $\alpha = \alpha' a p(x)$, where $a \in A$ and $p(x)$ is a point trajectory at x .

By the induction hypothesis, we know that $\alpha'.lstate \in I$.

By invoking the *transition closure*, we obtain $x \in I$.

Case 2: $\alpha = \alpha' \tau$, where τ is a trajectory of $\mathcal{A}$ and $\tau.lstate = x$

By the *induction hypothesis*, $\alpha'.lstate \in I$ and by

invoking the *trajectory closure*, we deduce that $\tau.lstate = x \in I$

An application

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$

trajectories:

Loc1

evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “stays above ground”

$I_0: x \geq 0 \equiv \{ \mathbf{u} \in \text{val}(\{x, v\}) \mid \mathbf{u}[x \geq 0] \}$

Applying Theorem 7.1:

- Consider any initial state $\mathbf{u} \in \Theta; \mathbf{u}[x = h] \geq 0$
 - $\mathbf{u} \in I_0$
- Consider any transition $\mathbf{u} \rightarrow_{\text{bounce}} \mathbf{u}'$
 - From precondition we know $\mathbf{u}[x = 0]$; from effect we know $\mathbf{u}'.x = \mathbf{u}.x$ therefore $\mathbf{u}'[x = 0] \geq 0$
 - $\mathbf{u}' \in I_0$
- Consider any trajectory $\tau \in T$
 - From mode invariant we know that for $\forall t \in \tau.\text{dom}, \tau(t)[x \geq 0]$
 - It follows that $\tau.\text{lstate}[x \geq 0]$
- What part of Bouncingball was used ? What could be changed?

An application

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$

trajectories:

Loc1

evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “stays above ground and below h”

$$I_h: h \geq x \geq 0$$

Applying Theorem 7.1:

- Consider any initial state $\mathbf{u} \in \Theta$; $\mathbf{u}[x = h]$
 - $\mathbf{u} \in I_h$
- Consider any transition $\mathbf{u} \rightarrow_{\text{bounce}} \mathbf{u}'$
 - From precondition we know $\mathbf{u}[x = 0]$; from effect we know $\mathbf{u}'.x = \mathbf{u}.x$ therefore $\mathbf{u}'[x = 0]$
 - $\mathbf{u}' \in I_h$
- Consider any trajectory $\tau \in T$
 - From mode invariant and inductive hypothesis we know that for $\forall t \in \tau.\text{dom}, \tau(t)[x \geq 0 \text{ and } \tau(0)[x \in [0, h]$ and that τ is a solution of $d(x) = v; d(v) = -g$
 - Is this adequate to infer $\tau.\text{lstate} \in I_h$?

Strengthened invariant

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

k: Nat := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$; $k := k + 1$

trajectories:

Loc1

evolve $d(x) = v$; $d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “stays above ground and below h”

$$I_v: v^2 - 2g(hc^{2k} - x) = 0$$

Applying Theorem 7.1:

- Consider any initial state $u \in \Theta$; $u[x = h; u[k = 0$
 - $u \in I_v$
- **Exercise:** Finish the rest

Summary

- Theorem 7.1 gives a sufficient condition for proving **inductive** invariants
- Not all invariants are inductive
- We often have to **strengthen** invariants to make them inductive
- Read examples in Chapter 7

Floyd-Hoare Proofs

The core idea of inductive invariants dates back to program analysis technique called **Floyd-Hoare logic (1969)** which gives a set of rules for deducing correctness of automata, programs

The logic is built on **Hoare triples**, which describes how the execution of a statement (or line of code) changes the state of the automaton:

$P \ c \ Q$ where

- P and Q are predicates on the program variables and are called the **precondition** and **postcondition**
- c is a statement describing program variable change

The triple implies that when the precondition P is met, execution of c establishes the postcondition Q

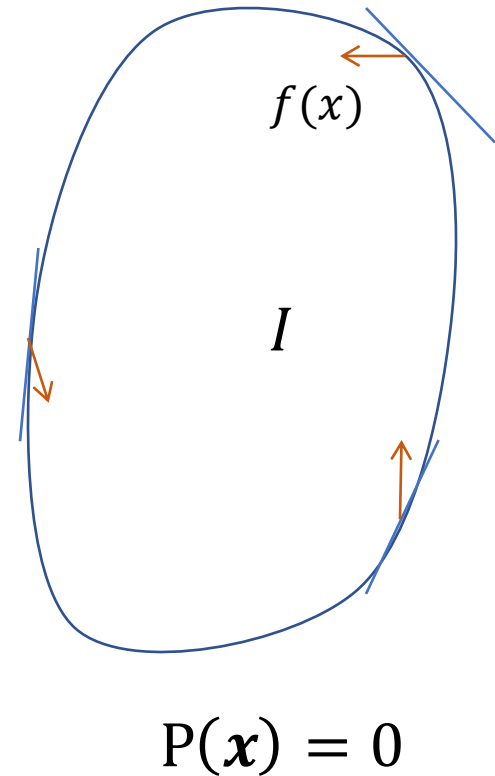
Sub-tangential conditions. Checking trajectory conditions without solving ODEs [Bhatia and Szegö 67]

How to check trajectory closure: for any $\tau \in \mathbf{T}$ if $\tau.fstate \in I$ then $\tau.lstate \in I$?

Use a Lyapunov function for each mode. How? Exercise.

Lemma. Consider the ODE $\dot{x} = f(x)$ for state variable x , describing $\mathbf{T}$. Let I be a compact set containing the initial set Θ . **Boundary of I is defined by $P(x) = 0$.** Then, I is an inductive invariant of the above ODE if for each x , *on the boundary* $P(x) = 0$, $f(x)$ is pointing inwards from the boundary. That is

$$\frac{\partial P(x)}{\partial x} \cdot f(x) \geq 0.$$



Related to the notions of Barrier Certificates and more general Control Barrier Functions.

Proving invariance using subtangential condition

Lemma. Consider $\dot{x} = f(x)$ and let $P: \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function such that $I = \{x | P(x) \leq 0\}$ is compact and contains. $\partial I := \{x | P(x) = 0\}$ is the boundary of I and P is differentiable over ∂I . If

Sub-tangential condition $\frac{\partial P(x)}{\partial x} \cdot f(x) \leq 0 \quad \forall x \in \partial I$.

Then I is an invariant

Proof. Fix $x(0) \in \Theta \subseteq I$. We must show $\forall t \geq 0, x(t) \in I$

Define $\psi(t) = P(x(t))$

$\psi(0) = P(x(0)) \leq 0 \quad [x(0) \in \Theta \subseteq I, P(x(0)) \leq 0]$

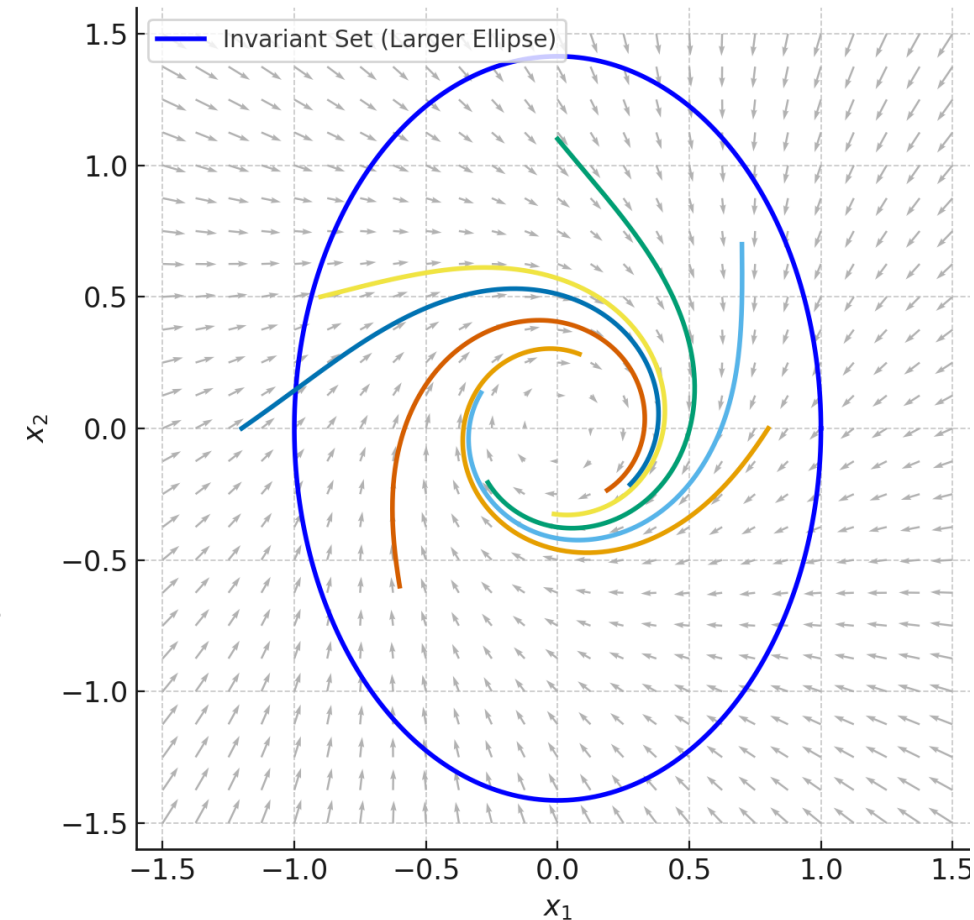
SFTC trajectory from $x(0)$ leaves I at some point t_1

That is, $\psi(t_1) = P(x(t_1)) > 0$

By continuity of ψ , there exists earliest $t^* \in (0, t_1)$ such that $\psi(t) \leq 0$ for $t \leq t^*$, $\psi(t^*) = 0 \Rightarrow x(t^*) \in \partial I$ and $\psi(t) > 0$ for some $t > t^*$

$$\dot{\psi}(t^*) = \frac{d}{dt} P(x(t)) \Big|_{t=t^*} = \frac{\partial P(x(t^*))}{\partial x} \cdot f(x(t^*)) \leq 0$$

$\psi(t^*)$ is nonincreasing and it is not possible for $\psi(t) > 0$ for $t > t^*$.



Lemma. Consider $\dot{x} = f(x)$ and let $P: \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function such that $I = \{x | P(x) \leq 0\}$ is compact and contains. $\partial I := \{x | P(x) = 0\}$ is the boundary of I and P is differentiable over ∂I . If

Sub-tangential condition $\frac{\partial P(x)}{\partial x} \cdot f(x) \leq 0 \quad \forall x \in \partial I$.

Then I is an invariant

Example.

$$\dot{x}_1 = x_2 - x_1(x_1^2 + x_2^2) \quad \dot{x}_2 = -x_1 - x_2(x_1^2 + x_2^2)$$

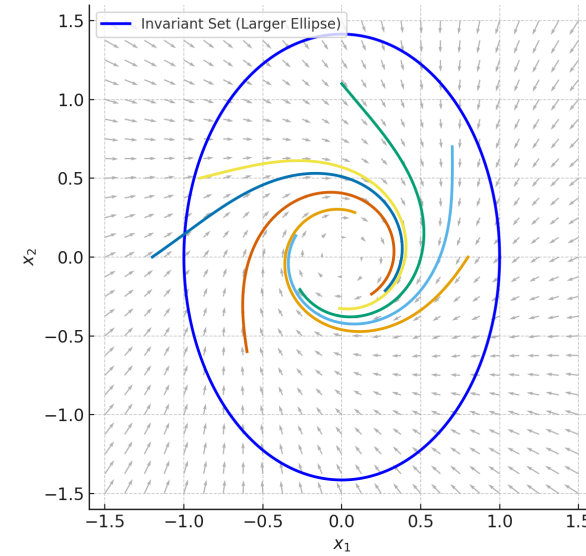
$$I = \{x \in \mathbb{R}^2 | P(x) \leq 0\} \text{ where } P(x) = 2x_1^2 + x_2^2 - 1$$

$$\nabla P = \frac{\partial P}{\partial x} = \begin{bmatrix} 4x_1 \\ 2x_2 \end{bmatrix}$$

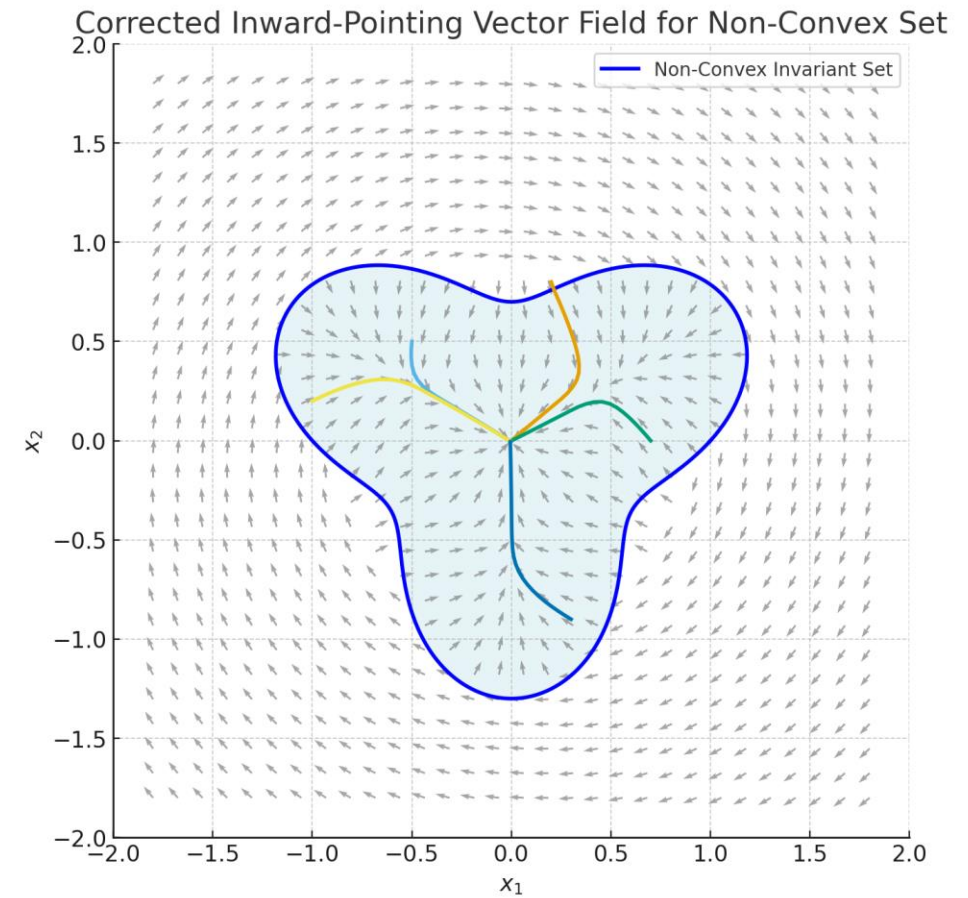
$$\begin{aligned} \nabla P \cdot f(x) &= \begin{bmatrix} 4x_1 \\ 2x_2 \end{bmatrix} \cdot [x_2 - x_1(x_1^2 + x_2^2) \quad -x_1 - x_2(x_1^2 + x_2^2)] \\ &= 4x_1(x_2 - x_1(x_1^2 + x_2^2)) + 2x_2(-x_1 - x_2(x_1^2 + x_2^2)) \\ &= 4x_1x_2 - 4x_1^2r^2 - 2x_2x_1 - 2x_2^2r^2 \\ &= 2x_1x_2 - 2r^2(2x_1^2 + x_2^2) \end{aligned}$$

$$\text{On the boundary } 2x_1^2 + x_2^2 = 1, \nabla P \cdot f(x) = 2x_1x_2 - 2r^2 = 2x_1x_2 - 2(x_1^2 + x_2^2)$$

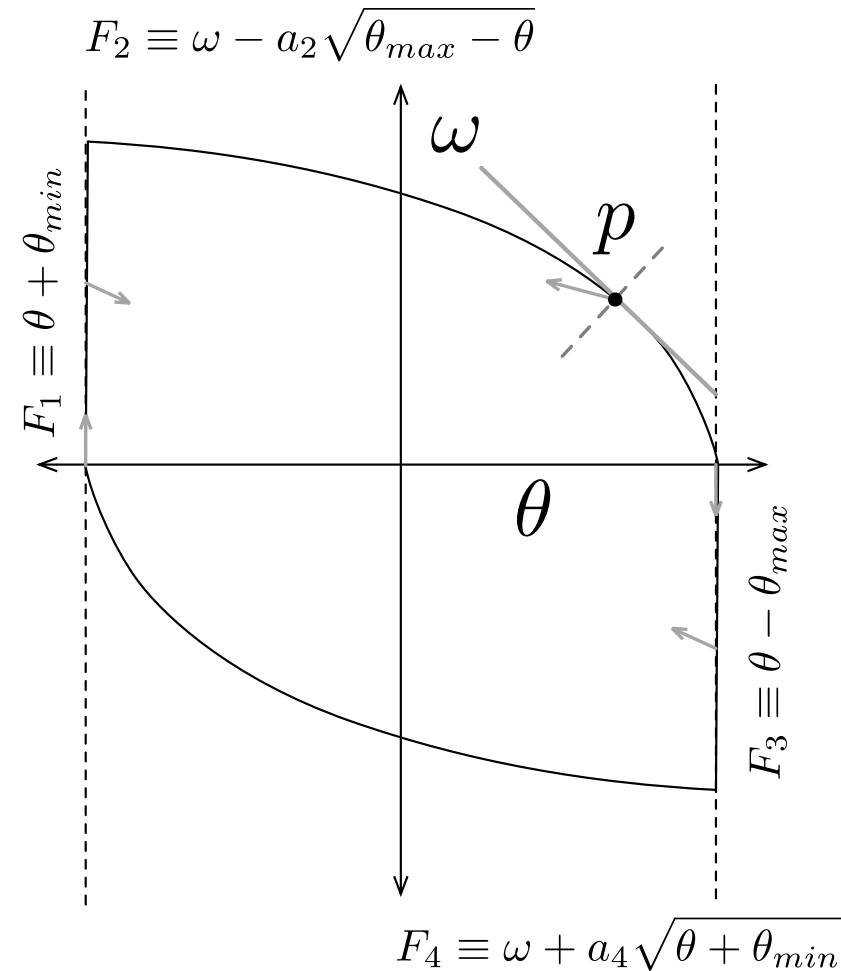
$$\text{Completing the square: } \nabla P \cdot f(x) = 2x_1x_2 - 2(x_1^2 + x_2^2) = -2\left(\frac{3}{4}x_1^2 + \left(x_2 - \frac{1}{2}x_1\right)^2\right) \leq 0$$



Another example



Checking sub-tangential condition in pieces



If the boundary is defined in pieces, for example, one for each mode, then check the condition for each piece

Summary

Interesting features in hybrid models

- Interaction between transitions, trajectories, and mode invariants can lead to extra sources of nondeterminism; transitions can be made urgent with mode invariants
- Mode switches can make the system unstable
- Zeno behavior

Inductive invariance proofs work with additional condition for trajectory closure

- Trajectory closure can be checked with subtangential condition, Lyapunov functions, Barrier certificates

